

ASSIGNMENT-3

Task – 1: File System Navigation.

SOLUTION-

1. / (Root Directory)

- It is the top-most directory in Linux.
- All other directories and files originate from this directory.

2. /home

- Stores personal files and folders of users.
- Each user gets a separate home directory.

3. /etc

- Contains system configuration files.
- Used by the operating system and services.

4. /var

- Stores variable data that changes frequently.
- Used for logs, cache, and mail files.

5. /tmp

- Used for temporary files.
- Files may be deleted automatically after reboot.

6. /usr

- Contains user programs and libraries.
- Most installed software is stored here.

7. /bin

- Contains essential Linux commands.
- Commands are available to all users.

8. /sbin

- Contains system administration commands.

- Mainly used by the root user.

9. /dev

- Contains device files representing hardware.
- Linux treats hardware devices as files.

10. /proc

- Virtual file system that provides process information.
- Created dynamically by the kernel.

Task – 2: File and Directory Operations

SOLUTION-

1. Create Directory Structure

Command

```
mkdir -p training/linux/day4
```

Verify

```
ls -R training
```

Output

```
training:
```

```
linux
```

```
training/linux:
```

```
day4
```

```
training/linux/day4:
```

2. Create 3 Text Files Inside day4

Command

```
touch training/linux/day4/file1.txt
```

```
touch training/linux/day4/file2.txt
```

```
touch training/linux/day4/file3.txt
```

Verify

```
ls training/linux/day4
```

Output

```
file1.txt
```

```
file2.txt
```

```
file3.txt
```

3. Copy One File to linux Directory

Command

```
cp training/linux/day4/file1.txt training/linux/
```

Verify

```
ls training/linux
```

Output

```
day4 file1.txt
```

4. Move One File to training Directory

Command

```
mv training/linux/day4/file2.txt training/
```

Verify

```
ls training
```

Output

```
file2.txt linux
```

5. Rename a File Using mv

Command

```
mv training/linux/day4/file3.txt training/linux/day4/renamed.txt
```

Verify

```
ls training/linux/day4
```

Output

```
file1.txt renamed.txt
```

6. Delete Directory Recursively Using rm -rf

Command

```
rm -rf training/linux/day4
```

Verify

```
ls training/linux
```

Output

```
file1.txt
```

Task – 3: File Content Commands

SOLUTION-

Step 1: Create a File with 20 Lines

Command

```
cat > sample.txt
```

Content

```
Linux is an operating system
```

```
Linux is open source
```

Ubuntu is a Linux distribution

Red Hat is used in enterprises

Linux supports multiuser environment

Linux provides security features

Shell is a command interpreter

Kernel is the core of Linux

DevOps uses Linux extensively

Docker runs well on Linux

Cloud computing uses Linux

Linux has many distributions

System administration is easier in Linux

Linux supports scripting

Bash is a popular shell

Linux manages processes efficiently

Linux supports networking

Linux is reliable and stable

Keyword appears here

Linux is widely used

Press Ctrl + D to save.

1. cat Command

Command

cat sample.txt

Purpose

Displays the entire content of a file.

Practical Use Case

A Linux administrator uses cat to quickly view configuration files and logs.

2. less Command

Command

less sample.txt

Purpose

Displays file content one screen at a time with forward and backward navigation.

Practical Use Case

Used to read large log files without opening them in an editor.

3. more Command

Command

more sample.txt

Purpose

Displays file content page by page.

Practical Use Case

Used when viewing long text files from the terminal.

4. head -5 Command

Command

head -5 sample.txt

Output

Linux is an operating system

Linux is open source

Ubuntu is a Linux distribution

Red Hat is used in enterprises

Linux supports multiuser environment

Purpose

Displays the first 5 lines of a file.

Practical Use Case

Used to verify the beginning of log or configuration files.

5. tail -5 Command

Command

tail -5 sample.txt

Output

Linux manages processes efficiently

Linux supports networking

Linux is reliable and stable

Keyword appears here

Linux is widely used

Purpose

Displays the last 5 lines of a file.

Practical Use Case

Used to monitor the latest entries in log files.

6. wc -l Command

Command

wc -l sample.txt

Output

20 sample.txt

Purpose

Counts the number of lines in a file.

Practical Use Case

Used to determine the size of log files and reports.

7. grep 'Keyword' Command

Command

grep "Keyword" sample.txt

Output

Keyword appears here

Purpose

Searches for specific text patterns in a file.

Practical Use Case

Used to find errors, warnings, or specific entries in log files.

8. sort Command

Command

sort sample.txt

Purpose

Sorts lines in alphabetical order.

Practical Use Case

Used to organize data and remove duplicates when combined with other commands.

Task – 4: Working with Commands – type, which, whereis

SOLUTION-

Objective

To understand how Linux identifies commands and to learn the difference between built-in commands, aliases, and external binaries.

1. ls Command

Commands

type ls

which ls

whereis ls

Sample Output

ls is aliased to 'ls --color=auto'

/usr/bin/ls

ls: /usr/bin/ls /usr/share/man/man1/ls.1.gz

Observation

ls is an alias that points to the external binary /usr/bin/ls.

2. cd Command

Commands

type cd

which cd

whereis cd

Sample Output

cd is a shell builtin

cd:

cd:

Observation

cd is a shell built-in command and does not have a separate executable file.

3. pwd Command

Commands

type pwd

which pwd

whereis pwd

Sample Output

pwd is a shell builtin

/usr/bin/pwd

pwd: /usr/bin/pwd /usr/share/man/man1/pwd.1.gz

Observation

pwd exists as both a shell built-in and an external binary.

4. grep Command

Commands

type grep

which grep

whereis grep

Sample Output

grep is /usr/bin/grep

/usr/bin/grep

grep: /usr/bin/grep /usr/share/man/man1/grep.1.gz

Observation

grep is an external binary stored in /usr/bin.

5. cat Command

Commands

type cat

which cat

whereis cat

Sample Output

cat is /usr/bin/cat

/usr/bin/cat

cat: /usr/bin/cat /usr/share/man/man1/cat.1.gz

Observation

cat is an external binary located in /usr/bin.

Purpose of Commands

type

- Displays how the shell interprets a command.
- Identifies whether the command is an alias, built-in command, or executable file.

which

- Displays the path of the executable command.
- Helps locate the command being executed.

whereis

- Displays the location of the binary file, source file, and manual page.
 - Useful for finding related command files.
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Task – 5: Advanced Keyboard Tricks Practice

SOLUTION-

1. Ctrl + C

Purpose

Terminates the currently running command or process.

Example

```
ping google.com
```

Press **Ctrl + C**

Result

The ping command stops immediately.

2. Ctrl + Z

Purpose

Suspends the currently running process and sends it to the background.

Example

```
sleep 100
```

Press **Ctrl + Z**

Result

```
[1]+ Stopped sleep 100
```

3. Ctrl + D

Purpose

Logs out of the current shell or signals End Of File (EOF).

Example

```
cat > file.txt
```

Type some text and press **Ctrl + D**

Result

The file is saved and the command exits.

4. Ctrl + L

Purpose

Clears the terminal screen.

Example

```
ls -l
```

Press **Ctrl + L**

Result

The screen is cleared while keeping the current session active.

5. Ctrl + A

Purpose

Moves the cursor to the beginning of the command line.

Example

```
echo Linux Administration
```

Press **Ctrl + A**

Result

The cursor moves to the start of the line.

6. Ctrl + E

Purpose

Moves the cursor to the end of the command line.

Example

```
echo Linux Administration
```

Press **Ctrl + E**

Result

The cursor moves to the end of the line.

7. Ctrl + R (Reverse Search)

Purpose

Searches previously executed commands from command history.

Example

Press:

Ctrl + R

Type:

ls

Result

(reverse-i-search)'ls': ls -l

8. Tab Completion

Purpose

Automatically completes command names, file names, and directory names.

Example

cd Do

Press **Tab**

Result

cd Downloads/

9. history Command

Purpose

Displays a list of previously executed commands.

Example

history

Sample Output

101 ls

102 pwd

103 mkdir demo

104 cd demo

105 touch file.txt

Result

Shows command history with line numbers.

Benefits of These Shortcuts

- **Ctrl + C** quickly stops unwanted processes.
- **Ctrl + Z** pauses a process without terminating it.
- **Ctrl + D** exits shells and saves input efficiently.
- **Ctrl + L** keeps the terminal clean and readable.
- **Ctrl + A** and **Ctrl + E** allow fast cursor movement.
- **Ctrl + R** helps find previously used commands quickly.
- **Tab Completion** reduces typing and prevents mistakes.
- **history** helps review and reuse previous commands.